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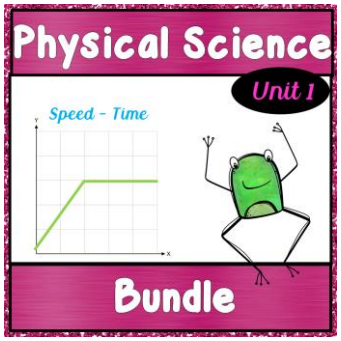
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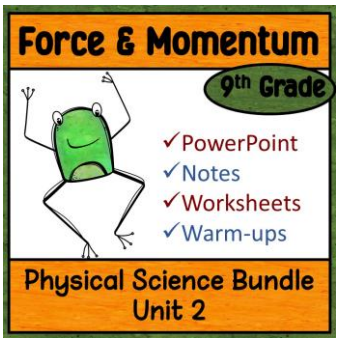
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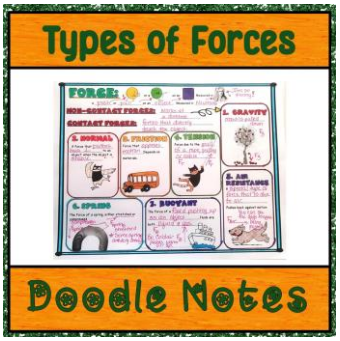
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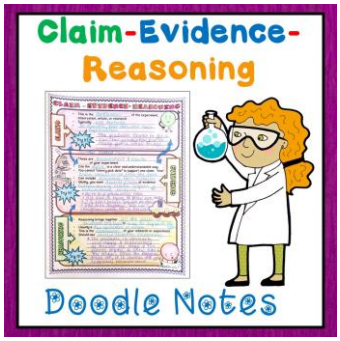
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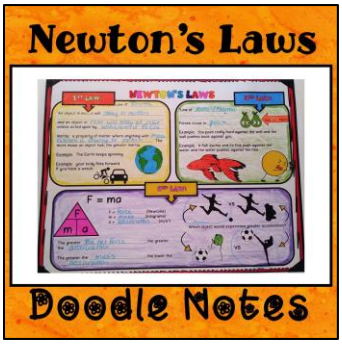












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Work, Energy, and Power Notes

I) I can identify 9 types of energy.

Energy: the ability to _____

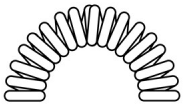
_____. Measured in _____



Chemical: Energy stored in _____or energy that is _____

_____

Examples: energy stored in food, batteries, plants, biomass, petroleum, and natural gas.



Elastic energy: energy from _____

Energy is stored when the object is _____

The energy is released when the object returns to its _____

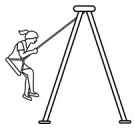
Examples: springs, trampolines, a bow (archery), bent diving board



Electrical Energy: energy from _____

Examples: lightening, static electricity, current electricity

Gravitational Energy: Energy due to _____



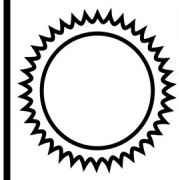
_____

Examples: lifting any object so that it can fall back to Earth. Such as roller coasters, or a boulder resting on a cliff.



Mechanical Energy: any kind of _____.

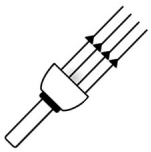
Examples: any movement such as cars, roller coasters, and walking. It also includes energy due to position such as boulders on a hill.



Nuclear Energy: energy from an atoms nucleus _____

_____Drawback: Fission and fusion make radioactive waste.

Examples: _____is splitting of an atom and is the type of atomic bomb dropped on Hiroshima and Nagasaki. _____ is atoms smashing together to form new elements and is what happens in the sun.



Radiant Energy: _____Energy from the entire _____

_____This is energy that can travel in a vacuum such as space.

Examples: radio waves, tv waves, microwaves, ultraviolet light, visible light, gamma waves, x-rays, and infrared.



Sound: energy that comes from a _____

_____(it cannot move in space). It moves through solids, liquids, and gases.



Thermal Energy: the energy from _____.

All matter is constantly moving except at _____

(-273 °C). Friction causes thermal energy as well.

Examples: the hotter something is, the greater the movement of molecules.

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2) I can describe and calculate potential energy.

Potential Energy: This is energy due to _____. The potential energy is greater the higher it is, and the more mass the object has.

$$PE = mgh$$

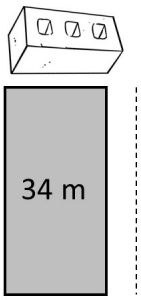
PE = _____

m = _____

g = _____

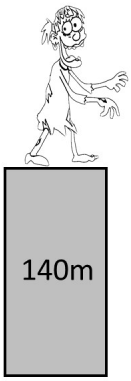
h = _____

Example 1: A cinderblock is teetering on top of a building that is 34 meters high. If the cinder block has a mass of 16 kg, find its potential energy.

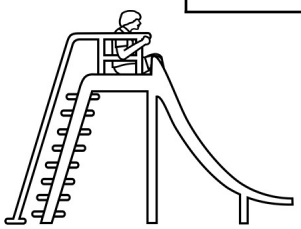


Define Variables	Write equation and show work	Answer w/ units
PE = m = g = h =		

Example 2: What is the mass of a zombie that is perched on top of a cliff that is 140 meters high and has potential energy of 96,000 J?



Define Variables	Write equation and show work	Answer w/ units
PE = m = g = h =		



Example 3: What is the height of the slide if a 38 kg kid has 560 J of potential energy at the top of the slide?

Define Variables	Write equation and show work	Answer w/ units
PE = m = g = h =		

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3) I can describe and calculate kinetic energy.

Kinetic energy: _____.

The object must be moving to have kinetic energy. It is dependent upon the mass of the object and the magnitude of the velocity.

$$KE = \frac{1}{2} mv^2$$

KE = _____

m = _____

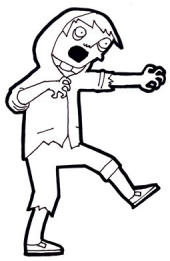
v = _____

Example 1: What is the kinetic energy of a car if it has a mass of 1,100 kilograms and is moving at 35 m/s?



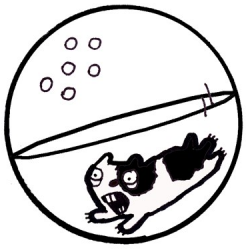
Define Variables	Write equation and show work	Answer w/ units
KE = m = v =		

Example 2: A zombie runs by at 12 m/s and has 8200 Joules of kinetic energy. What is the zombie's mass?



Define Variables	Write equation and show work	Answer w/ units
KE = m = v =		

Example 3: A zombie in a hamster ball goes rolling by. (don't ask questions. We don't know why). What is the velocity of the zombie hamster if it has a mass of 0.5 kg and 200 Joules of kinetic energy.



Define Variables	Write equation and show work	Answer w/ units
KE = m = v =		

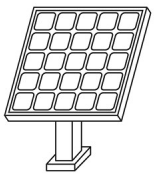
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4) I can describe the conservation of energy and I can describe energy transfers between any of the 9 energy types.

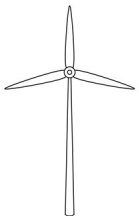
Conservation of Energy: energy is neither _____
It is converted from _____

You can describe the flow of energy between the types of energy.

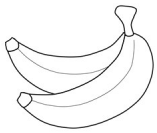
Example 1: Describe the energy transformations that happen with a solar panel.

_____ →  → _____
_____ _____

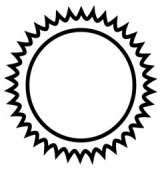
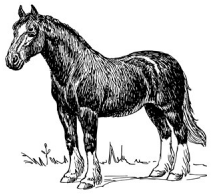
Example 2: Describe the energy transformations that happen with a wind turbine.

_____ _____ _____
_____ →  _____ _____ → _____ _____
_____ _____ _____

Example 3: Describe the energy transformations of a person that eats a banana and goes on a run.

_____ →  → _____  → _____
_____ _____ _____

Example 4: Energy starts in the sun and is transmitted to an apple that is eaten by a horse.

 → _____  → 
_____ _____ _____
_____ _____ _____

Example 5:. A car is filled with gasoline and races off. The car honks its horn and creates friction with the road.

 →  →  + friction w/ road
_____ _____ _____ _____
_____ _____ _____ _____

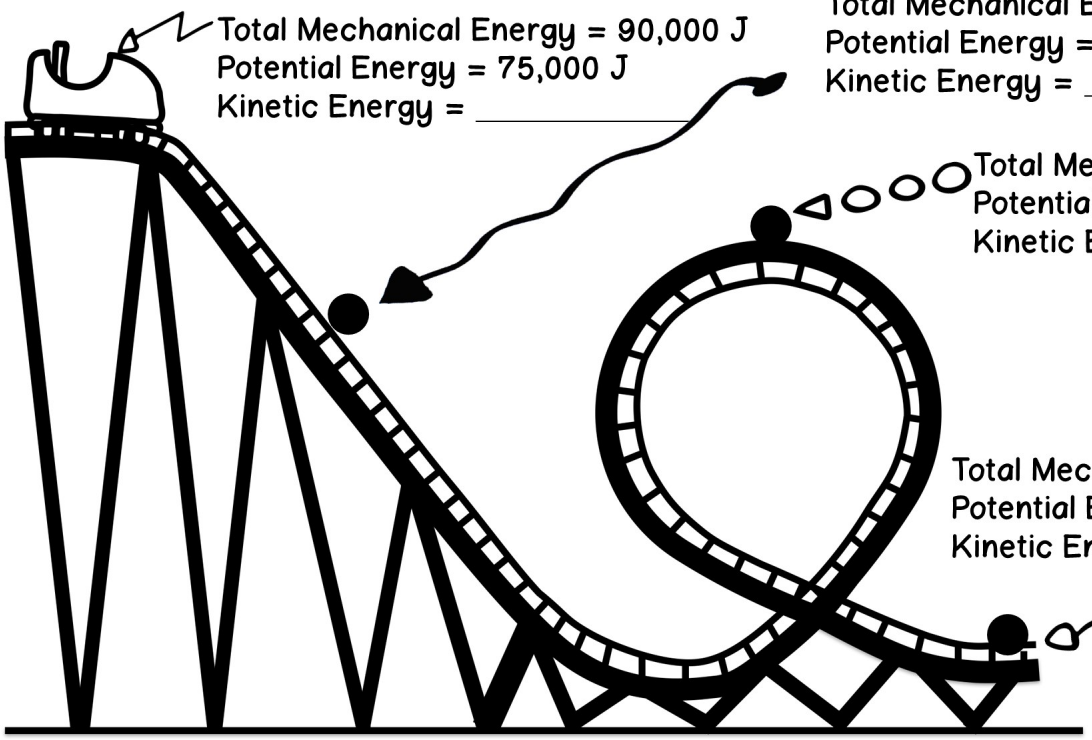
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5) I can describe and calculate the transformation of potential and kinetic energy (total mechanical energy) in a frictionless system.

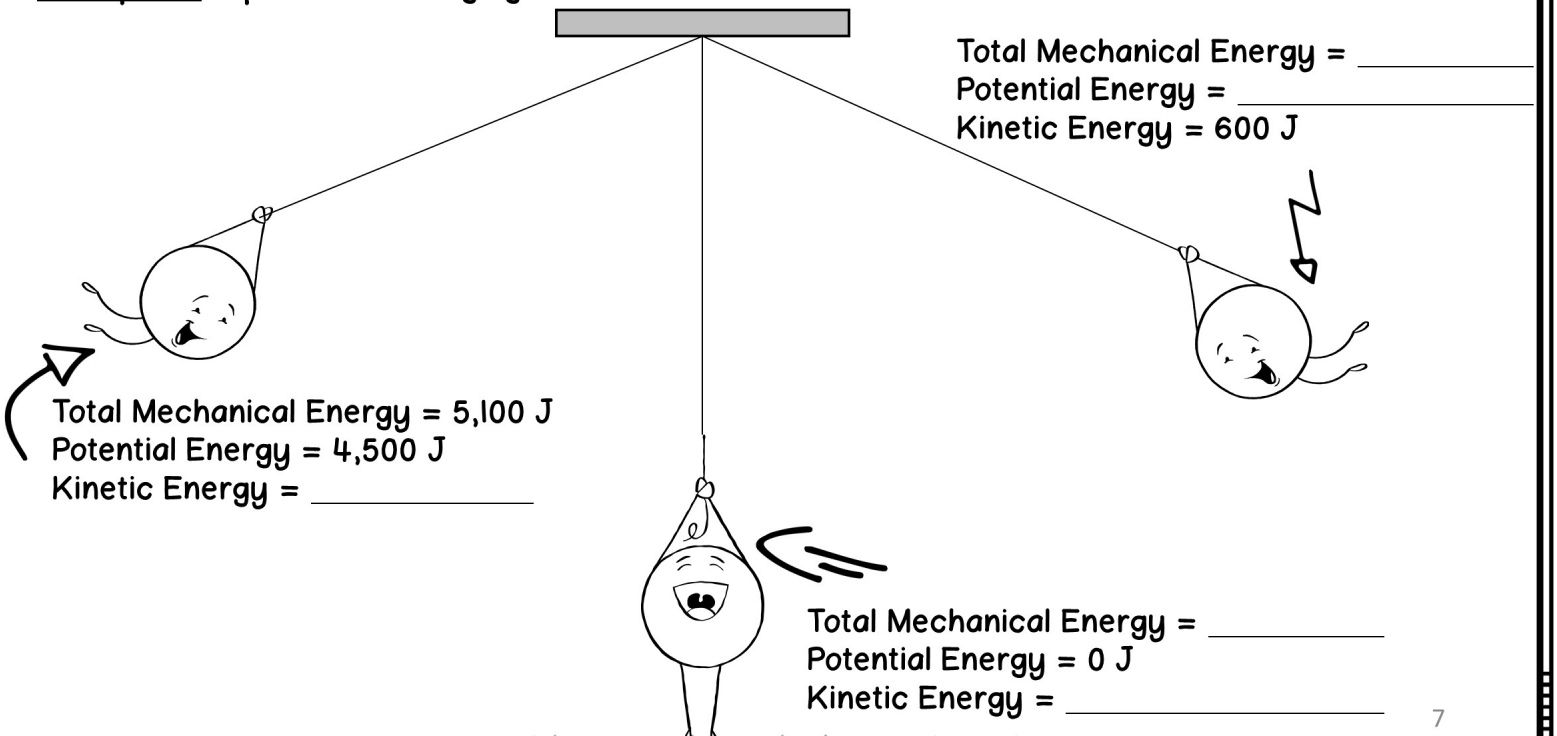
Mechanical Energy: is both _____(kinetic energy) and _____ _____(potential energy). In a frictionless system, energy is converted back and forth.

- You can add kinetic plus potential at each stage to track the energy
- Total mechanical energy: the _____at each point of movement.

Example 1: A rollercoaster



Example 2: A pendulum swinging back and forth.



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6) I can describe and calculate work.

Work: The transfer of _____.

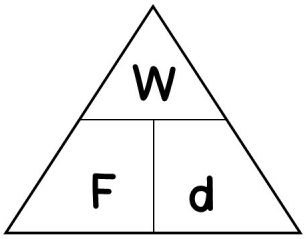
Measured in _____

W = Fd

W = _____

F = _____

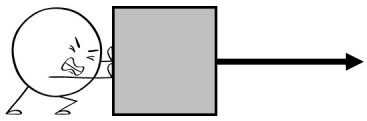
d = _____



For work to happen, there are two conditions that must occur:

1. The object must move _____
2. The force and the object must be _____

Example A: Dot Dude pushes a block and accelerates it from rest to 2 m/s to the right

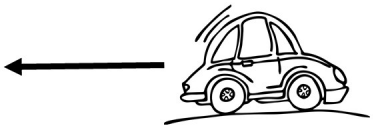


This is an example of

_____

_____

Example B: A car slows down from 20 m/s to 10 m/s.

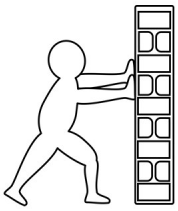


This is an example of

_____

_slows from 20 m/s to 10 m/s_____

Non-Example A: You push really, really hard against a brick wall. You get sweaty and tired. The wall doesn't move.

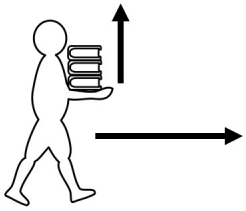


This is an example of _____

_____.

Why? The wall didn't move.

Non-Example B: You carry a super heavy stack of books (25 kg worth) across the room (10 meters). Your arms are tired afterwards.



This is a tricky example of _____

_____.

Why? Because the arms are pulling up and the walking is going forward. The force and the distance are _____

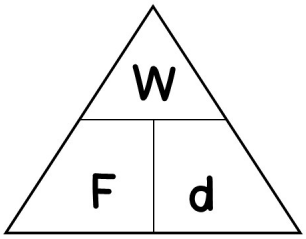
_____

Example 1: You use 35 Newtons to push a lawn mower 15 meters. What is the work done on the lawn mower?

Define Variables	Write equation and show work	Answer w/ units
W = F = d =		

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$$W = Fd$$



Example 2: You lift your backpack and use 25 Joules of energy. If it took you 59 Newtons of force, how far did lift your backpack?

Define Variables	Write equation and show work	Answer w/ units
W = F = d =		

Example 3: How much work is done to stop a car with a force 27,000 N of force over a distance 56 meters?

Define Variables	Write equation and show work	Answer w/ units
W = F = d =		

Example 4: An elevator lifts 12 people up 77 meters with 640,000 Joules of work. What is the force exerted on the elevator?

Define Variables	Write equation and show work	Answer w/ units
W = F = d =		

Example 5: A person is hauling their taco stand and it takes 3,500 Joules of work to stop the taco stand. What force was exerted on the taco stand if it took 1.5 meters to stop the motion?

Define Variables	Write equation and show work	Answer w/ units
W = F = d =		

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7) I can describe and calculate power.

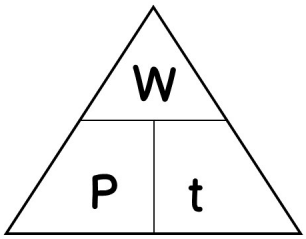
Power: how _____

$$P = \frac{W}{t}$$

P = _____

W = _____

t = _____



Note: They can combine the power equation and the work equation for a two-step problem. Or you can use this modified equation:

$$W = Fd \quad + \quad P = \frac{W}{t} \quad = \quad P = \frac{Fd}{t}$$

Example 1: Machine A does 5000 Joules of work in 5 seconds. Machine B does 5000 Joules of work in 25 seconds. Find both powers. Which machine has more power?

Define Variables	Define Variables	Write equation and show work	Answer w/ units
P = W = t =	P = W = t =		

Example 2: How much power is created by lifting with 350 Newtons over 12 meters, if it took 5.2 seconds to accomplish this?

Define Variables	Write equation and show work	Answer w/ units
P = W = F = d = t =		

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Example 3: A machine uses 4,500 Watts of power to do 25,000 Joules of work. How much time did the machine take to do this job?

Define Variables	Write equation and show work	Answer w/ units
P = W = F = d = t =		

Example 4: You do 540 Watts of power in 45 seconds by walking on some stairs. How high were the stairs if it took you 750 Newtons of force to move?

Define Variables	Write equation and show work	Answer w/ units
P = W = F = d = t =		

Example 5: You do chin-ups and move your body with 525 Newtons of force over a distance of 0.25 meters. It takes you 1.1 seconds. What power do you exert?

Define Variables	Write equation and show work	Answer w/ units
P = W = F = d = t =		

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